1 Solution — GIAM 1.6 Problem 3

1.1 Problem

If a number x is even, it is easy to show that its square x^2 is even. The lemma that went unproved in this section asks us to start with a square x^2 that is even and deduce that the unsquared number x is even. Perform some numerical experimentation to check whether this assertion is reasonable. Can you give an argument that would prove it?

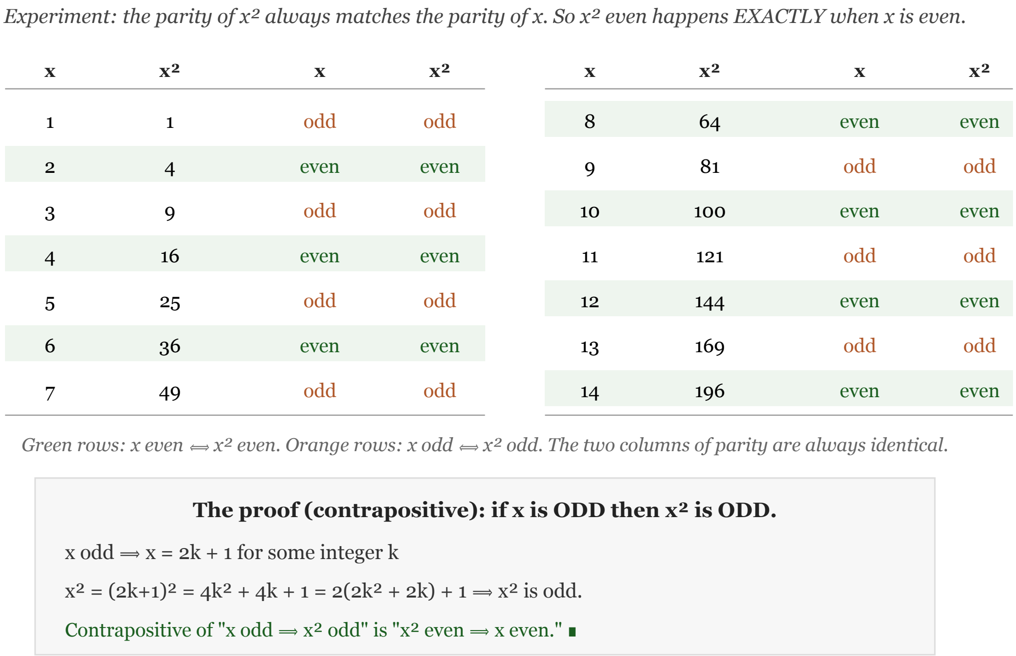
1.2 The Claim (Lemma 1.6.2)

$$\boxed{\text{If } x^2 \text{ is even, then } x \text{ is even.}}$$

Equivalently: **the parity of x^2 always equals the parity of x** — a square is even exactly when its root is even.

1.3 Numerical Experimentation

Tabulate x and x^2 for the first several integers and record the parities:



Experiment: the parity of x^2 always matches the parity of x. So x^2 even happens EXACTLY when x is even.

x	x^2	x	x^2
1	1	odd	odd
2	4	even	even
3	9	odd	odd
4	16	even	even
5	25	odd	odd
6	36	even	even
7	49	odd	odd

x	x^2	x	x^2
8	64	even	even
9	81	odd	odd
10	100	even	even
11	121	odd	odd
12	144	even	even
13	169	odd	odd
14	196	even	even

Green rows: x even $\Longrightarrow$ x^2 even. Orange rows: x odd $\Longrightarrow$ x^2 odd. The two columns of parity are always identical.

The proof (contrapositive): if x is ODD then x^2 is ODD.

x odd $\Longrightarrow$ x = 2k + 1 for some integer k

$x^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 \Longrightarrow x^2$ is odd.

Contrapositive of "x odd $\Longrightarrow$ x^2 odd" is "x^2 even $\Longrightarrow$ x even." ∎

Figure 1.1: For x from 1 to 14: the parity of x-squared always matches the parity of x. Even x (2,4,6,8,10,12,14) give even squares (4,16,36,64,100,144,196); odd x (1,3,5,7,9,11,13) give odd squares (1,9,25,49,81,121,169). So x-squared is even exactly when x is even. The proof by contrapositive: x odd means x = 2k+1, so x-squared = $4k^2+4k+1$ = $2(2k^2+2k)+1$, which is odd; the contrapositive of ‘x odd $\Longrightarrow$ x-squared odd’ is ‘x-squared even $\Longrightarrow$ x even’.

x	x^2	x	x^2	conclusion
2	4	even	even	even $\to$ even ✓
3	9	odd	odd	odd $\to$ odd ✓
6	36	even	even	even $\to$ even ✓
7	49	odd	odd	odd $\to$ odd ✓
11	121	odd	odd	odd $\to$ odd ✓
12	144	even	even	even $\to$ even ✓

Table 1.1

In every case an even square comes from an even x, and we *never* see an even square produced by an odd x. The assertion looks entirely reasonable — no counterexample appears.

1.4 The Proof — by Contrapositive

Numerical checking builds confidence but proves nothing (there are infinitely many integers). For a proof, the direct route — "assume x^2 even, show x even" — is awkward, so we prove the logically equivalent **contrapositive**:

$$\text{if } x \text{ is } \textbf{odd}, \text{ then } x^2 \text{ is } \textbf{odd}.$$

Suppose x is odd. Then $x = 2k + 1$ for some integer k. Squaring,

$$x^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2\underbrace{(2k^2 + 2k)}_{\text{an integer}} + 1,$$

which is $2 \cdot (\text{integer}) + 1$ — odd. So an odd x always has an odd square.

The statement "x odd $\Rightarrow x^2$ odd" and its contrapositive "x^2 even $\Rightarrow x$ even" are logically the same statement, so we have proved the lemma. ■

1.5 Remark — Why the Contrapositive Instead of a Direct Proof

A direct attack ("$x^2 = 2m$, therefore $x = ?$") stalls: from $x^2 = 2m$ there is no algebraic way to *extract* x and read off its parity. The contrapositive flips the problem to start from x odd, where we have an explicit handle — $x = 2k + 1$ — that we can square and simplify directly. This is the payoff of contraposition: $P \Rightarrow Q$ and $(\neg Q) \Rightarrow (\neg P)$ are equivalent, so we are free to prove whichever direction gives us something concrete to compute with. Here, "not even $=$ odd" hands us a formula, so the contrapositive is the easy side.

1.6 Remark — Experimentation vs. Proof

The problem deliberately pairs *experimentation* with *proof* to make a methodological point. The table checks 14 cases and finds no counterexample — genuinely useful for spotting whether a claim is even *plausible* before investing in a proof. But no finite table can establish a statement about *all* integers: a pattern that holds for the first billion cases can still fail later (many famous conjectures do). The algebraic argument, by contrast, covers every odd x at once through the single symbol k. Experimentation suggests; only proof settles.

1.7 Remark — The Same Fact via Prime Factorization

There is a second proof, straight from Problem 3 of the previous section. In the factorization of x, let the exponent of the prime 2 be e. Squaring *doubles* every exponent, so the exponent of 2 in x^2 is $2e$. Now x is even $\iff e \geq 1$, and x^2 is even $\iff 2e \geq 1 \iff e \geq 1$. The two conditions are identical, so x^2 even $\iff x$ even. This viewpoint also explains the general pattern: for any prime p, $p \mid x^2 \iff p \mid x$, because a prime dividing a square must divide the base — the exact fact the $\sqrt{2}$ proof needs.

1.8 Remark — This Is the Missing Gear in the $\sqrt{2}$ Proof

This lemma is not an idle curiosity; it is the load-bearing step in the section's proof that $\sqrt{2}$ is irrational. There one writes $\sqrt{2} = a/b$ in lowest terms, squares to get $a^2 = 2b^2$, and concludes a^2 is even. To continue — "so a is even" — one needs *exactly* this lemma. Then $a = 2c$ gives $4c^2 = 2b^2$, i.e. $b^2 = 2c^2$, so b^2 is even and (lemma again) b is even. But now a and b are both even, contradicting "lowest terms" ($\gcd(a, b) = 1$). The whole irrationality argument hinges on being able to pass from "x^2 even" to "x even," which is why the text flagged it as the lemma that had gone unproved.